THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 65 degrees	i = 90 - 25 = 65.
2	**(c)** 45 degrees	Shadow = height.
3	**(a)** small fish	Third level.
4	**(d)** less than 1 per cent of the radius	35/6400 is about 0.5 per cent.
5	**(b)** 20 cm	R = 2f.
6	**(a)** 354	12 x 29.5.
7	**(c)** oxygen	Photosynthesis.
8	**(b)** 105 L	21/100 x 500.
9	**(a)** principal focus	Focus.
10	**(a)** west after sunset	Young Moon.
11	**(c)** organic waste rots in it	Decomposition uses oxygen.
12	**(c)** 3 per cent	Most is salty.
13	**(d)** 40 degrees	i + r = 80.
14	**(d)** to the south of the observer	Shadow is opposite to the Sun.
15	**(d)** foxes are removed	Fewer predators.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	i = 60 degrees (1); Total angle = 120 degrees (1).	1+1
17	Shadow = height (1); Angle = 45 degrees (1).	1+1
18	Energy decreases at each trophic level (1); Less energy is available to support them (1).	1+1
19	The pressure is enormous (1); It keeps the iron-nickel solid (1).	1+1
20	Convex mirror diverges rays (1); A virtual, diminished image is formed (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	New i = 50 degrees (or 30) (1); Reflected ray turns by twice 10 (1); = 20 degrees (1).	1+1+1
22	9 am: long, west (1); 12 noon: shortest (1); 3 pm: long, east and symmetrical with 9 am (1).	1+1+1
23	Algae: producer (1); Zooplankton: primary consumer (1); Fish and heron: secondary and tertiary consumers (1).	1+1+1
24	Radius = crust + mantle + core (1); Core = 6400 - 35 - 2900 (1); = 3465 km (1).	1+1+1
	OR: Wide field of view (1); Image erect (1); Objects appear smaller and farther (1).	1+1+1
25	Gregorian: solar, 365 or 366 days (1); Islamic: lunar, about 354 days (1); Hindu: lunisolar with extra months (1).	1+1+1
	OR: Deer population rises (1); Overgrazing destroys vegetation (1); Ecosystem degraded (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Mirror, white paper, pins, protractor (1); Draw normal and incident ray (1); Fix pins on reflected ray (1); Measure i and r (1); i = r, all in one plane (1).	1+1+1+1+1
	OR: Fix a vertical stick (1); Mark the tip of the shadow every 10 minutes (1); Shortest shadow is local noon (1); It lies along the north-south line (1); Compare with clock time (1).	1+1+1+1+1
27	Grass to grasshopper to frog to snake to hawk (1); Grass to mouse to snake to hawk (1); Hawk eats mice and snakes (1); Without snakes mice and frogs increase (1); Hawks have fewer prey; grass is overgrazed (1).	1+1+1+1+1

	OR: Crust: 5 to 70 km, solid rock (1); Mantle: about 2900 km, hot semi-solid rock (1); Outer core: liquid iron-nickel (1); Inner core: solid iron-nickel (1); Seismic waves reveal it (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** a thick CO2 atmosphere	Runaway greenhouse.
2	**(d)** convex mirror	A convex mirror gives a wide field of view.
3	**(d)** 11 days	365 - 354.
4	**(a)** global warming	Greenhouse effect.
5	**(d)** extremely high pressure	Pressure keeps it solid.
6	**(c)** virtual, erect and smaller	Always.
7	**(a)** sunset	Opposite to the Sun.
8	**(c)** increase	Fewer predators.
9	**(c)** 156 L	78/100 x 200.
10	**(c)** 4 m/s	Relative speed.
11	**(b)** 15 days	Half a month.
12	**(c)** producer level	Energy decreases at each step.
13	**(c)** 30	3 per cent.
14	**(a)** virtual, erect and enlarged	Magnifying mirror.
15	**(a)** 33 days	3 x 11.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Lunar month is 29.5 days (1); 12 x 29.5 = 354 (1).	1+1
17	Kills useful insects as well (1); Passes along the food chain (1).	1+1
18	Troposphere is heated from the ground (1); Stratosphere is heated by ozone absorbing UV (1).	1+1
19	Rays meet at the focus (1); An inverted real image forms (1).	1+1
20	Heights are in the ratio of shadows (1); Height = 12 m (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Earth has suitable temperature and air pressure (1); Moon has no atmosphere (1); Water would evaporate or freeze (1).	1+1+1
22	Concave: converges at the focus (1); Convex: diverges, appears to come from the focus behind the mirror (1); Different effects of curvature (1).	1+1+1
23	North hemisphere tilts toward the Sun (1); Sun is above horizon for longer (1); Day is longer in summer (1).	1+1+1
24	Collects rain from roofs (1); Recharges groundwater (1); Reduces water scarcity (1).	1+1+1
	OR: Air molecules scatter blue light more (1); Blue reaches our eyes from all directions (1); In space the sky is black (1).	1+1+1
25	Polished surface gives regular reflection (1); Rough surface scatters (1); Image forms only in regular reflection (1).	1+1+1
	OR: Mark the shortest shadow (1); It lies along the north-south line (1); Shadow points to the north in India (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Minimum height = 0.9 m (1); Top edge at half the distance of head and eyes (1); Bottom edge at half the distance of feet and eyes (1); Independent of distance from mirror (1); Mirror length is half (1).	1+1+1+1+1
	OR: Latitude equals the altitude, so 25 degrees North (1); Find the Big Dipper (1); Extend the pointer stars (1); Five times the gap (1); Pole Star found (1).	1+1+1+1+1

27	Pond: algae, fish (1); Grassland: grass, herbivores (1); Decomposers present in both (1); Abiotic differs (1); Food chain examples (1).	1+1+1+1+1
	OR: CFCs from refrigerators (1); Release chlorine in the stratosphere (1); UV reaches Earth (1); Montreal Protocol (1); Layer slowly recovering (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** burning of fossil fuels and deforestation	Human activity.
2	**(a)** is farther from the Sun and has a thin atmosphere	Weak greenhouse.
3	**(b)** at the focus	Parallel rays.
4	**(c)** sunrise	Near the Sun.
5	**(b)** stray dogs and diseases	Carcasses remain.
6	**(b)** seismic waves of earthquakes	Waves travel through layers.
7	**(a)** 2 m	2 x 1.0.
8	**(a)** 15 January	Half a lunar month.
9	**(c)** more than one trophic level	Omnivore.
10	**(c)** air is thinner with lower oxygen amount per breath	Pressure falls.
11	**(b)** principal focus	Focus.
12	**(d)** Adhik maas	Corrects the gap.
13	**(b)** stomata	Transpiration.
14	**(c)** fossil groundwater in deep layers	Replenished very slowly.
15	**(d)** convex lens of small focal length	Closer to object than f.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Not biodegradable (1); Animals swallow and get entangled (1).	1+1
17	Air is thin at height (1); Less oxygen in each breath (1).	1+1
18	Gives an enlarged, erect image when close (1); Shows teeth in detail (1).	1+1
19	It is below the northern horizon (1); Because it lies above the northern axis (1).	1+1
20	Energy passes from the Sun to producers and then to consumers (1); Energy lost as heat cannot be used again by producers (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Frogs lose food and decrease (1); Snakes get less food from frogs (1); Plants may increase (1).	1+1+1
22	Earth was once molten (1); Dense iron sank to the centre (1); Lighter rocks rose to the crust (1).	1+1+1
23	Reflected at 30 degrees from the first mirror (1); Strikes second at 60 degrees to its normal (1); Emergent ray is antiparallel to the incident ray (1).	1+1+1
24	Altitude of the Pole Star equals latitude (1); Measured with an instrument (1); At the equator it is on the horizon (1).	1+1+1
	OR: Plants trap silt (1); Microbes break down pollutants (1); Clean water goes to rivers and groundwater (1).	1+1+1
25	Floods then drying of rivers (1); Sea level rises (1); Coastal areas submerged (1).	1+1+1
	OR: Lens forms an inverted real image (1); Brain interprets it as upright (1); Learned by the brain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	n = 360/60 - 1 = 5 (1); Images by repeated reflection (1); At 90 degrees n = 3 (1); Smaller angle gives more images (1); Parallel mirrors give infinite images (1).	1+1+1+1+1

	OR: New moon: Moon near Sun, rises with Sun (1); First quarter: rises at noon (1); Full moon: opposite the Sun, rises at sunset (1); Last quarter: rises at midnight (1); About 29.5 days cycle (1).	1+1+1+1+1
27	Producers (grass) form the broad base with most energy (1); Herbivores form the next level with less energy (1); Carnivores and top carnivores form higher, narrower levels (1); Energy is lost as heat at every step (1); So little energy remains for many more links (1).	1+1+1+1+1
	OR: Troposphere (to about 12 km): weather (1); Stratosphere (to about 50 km): ozone (1); Mesosphere (to about 85 km): meteors burn (1); Thermosphere: auroras and satellites (1); Exosphere: merges with space (1).	1+1+1+1+1

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